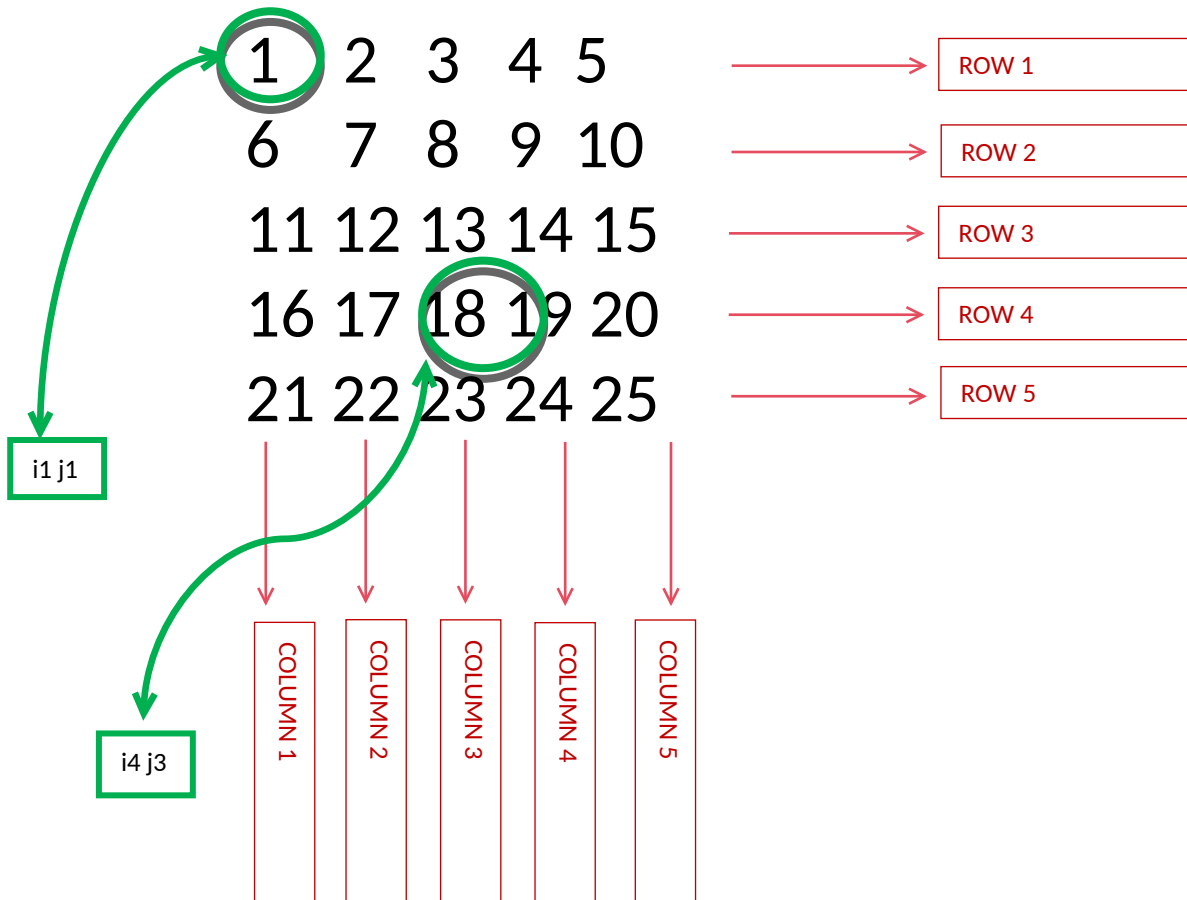


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

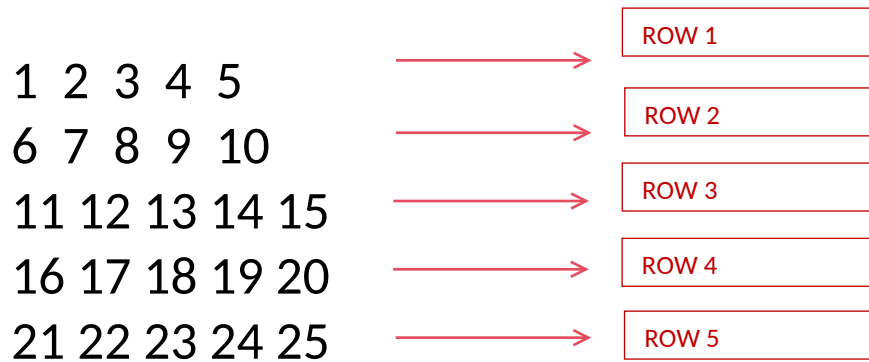
8. CONTINUOUS NUMBER SQUARE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a]square number pattern.
- number of rows is equal to the number of columns
- Unlike the Number Square and Same Number Square patterns, the values do not repeat row-wise or column-wise.
- Every row contains the sequence of numbers which are in continuously increasing order.
- For $n=5$



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

For $n = 5$

Row 1 : 1 2 3 4 5

Row 2 : 6 7 8 9 10

Row 3 : 11 12 13 14 15

Row 4 : 16 17 18 19 20

Row 5 : 21 22 23 24 25

Observations :-

- Number of rows = n
- Number of columns = n
- Every position is filled
- The numbers continue increasing regardless of row boundaries.
- No row starts again from 1.

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.
- and initialize a variable count
- for (i = 1 to n)

Step 3: For every row generated by the outer loop, traverse all columns of that row using the inner loop.

-The inner loop is responsible for generating columns.

- for (j = 1 to n)

Step 4: Print the current value of count and increment the counter

Step 5: After all numbers of the all columns of the current row are printed, move to the next line.

Step 6: Repeat the same process until all n rows have been completed.

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class ContinuousNumberSquare {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the square: ");
        int n = sc.nextInt();
        int count = 1;

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= n; j++) {
                System.out.print( count + " ");
                count++;
            }

            System.out.println();
        }

        sc.close();
    }
}
```

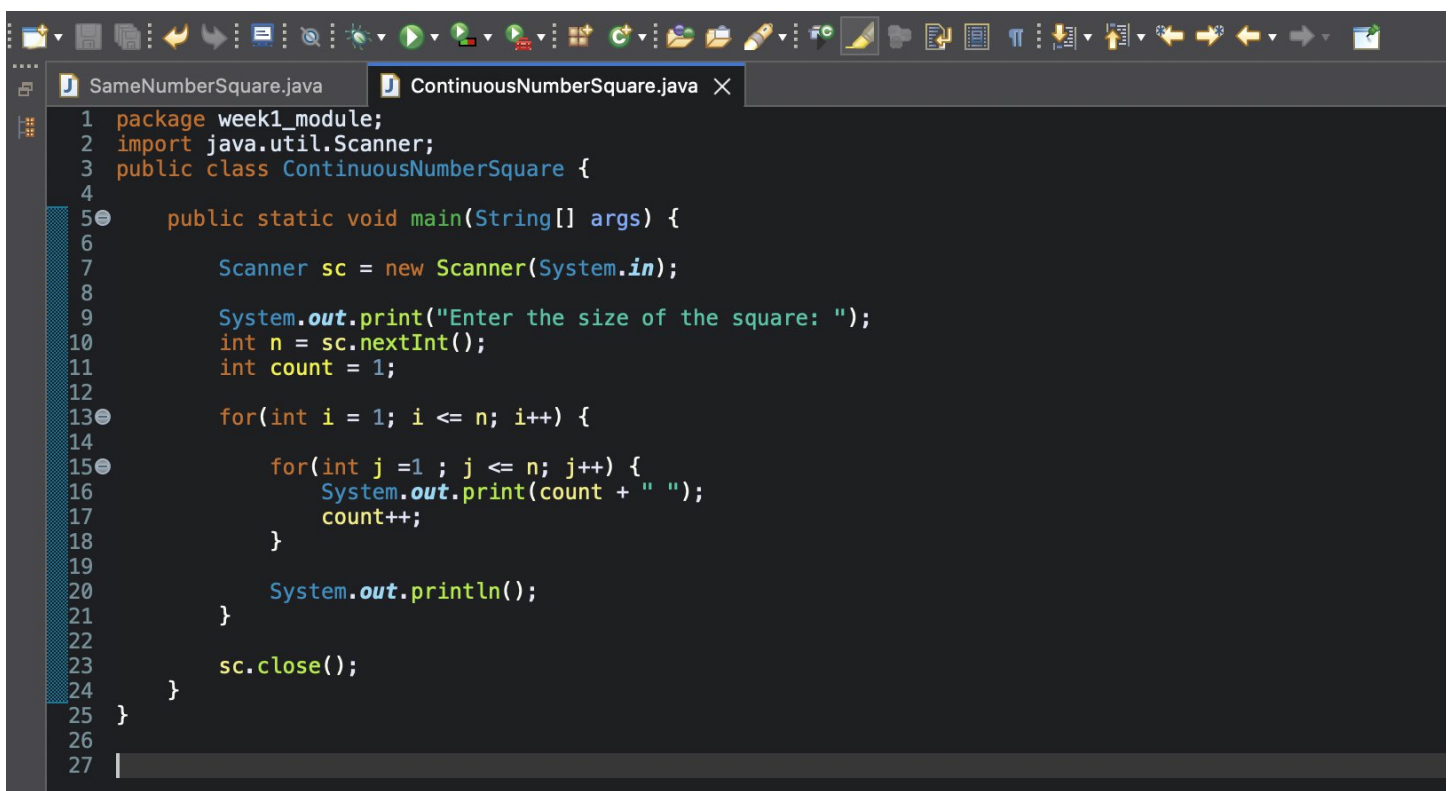
3. Program Output :-

Enter the size of the square: 5

```
1  2  3  4  5
6  7  8  9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
```

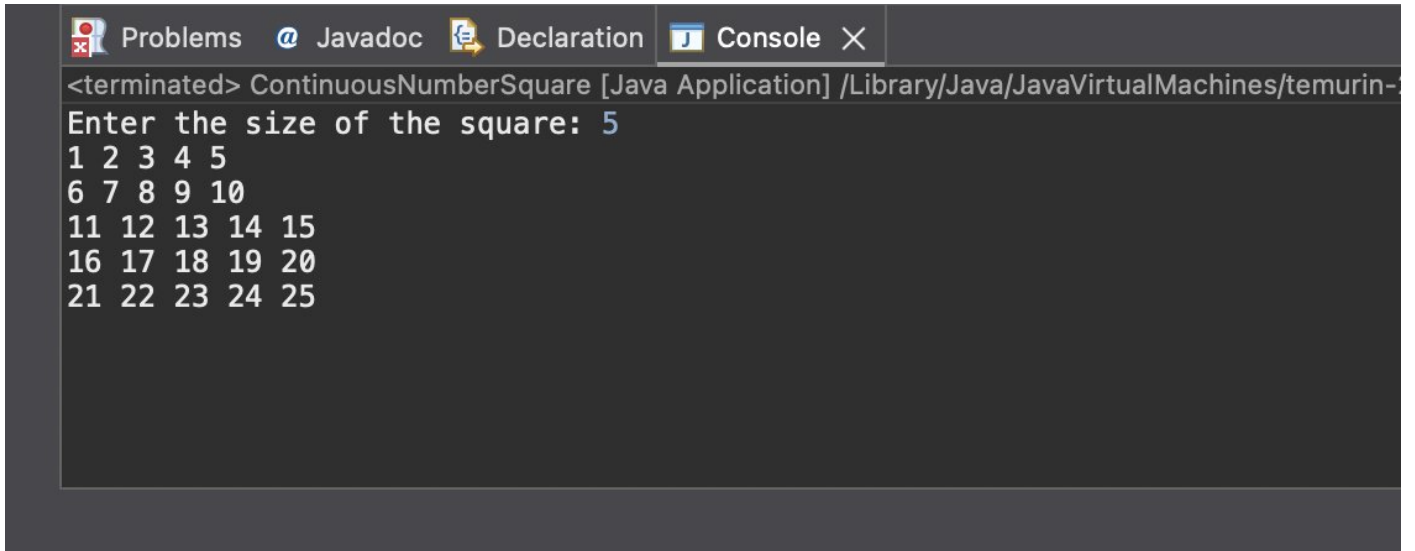
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class ContinuousNumberSquare {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the size of the square: ");
10        int n = sc.nextInt();
11        int count = 1;
12
13        for(int i = 1; i <= n; i++) {
14
15            for(int j = 1; j <= n; j++) {
16                System.out.print(count + " ");
17                count++;
18            }
19
20            System.out.println();
21        }
22
23        sc.close();
24    }
25 }
26
27 |
```

ii. Program's Output :-



```
<terminated> ContinuousNumberSquare [Java Application] /Library/Java/JavaVirtualMachines/temurin-  
Enter the size of the square: 5  
1 2 3 4 5  
6 7 8 9 10  
11 12 13 14 15  
16 17 18 19 20  
21 22 23 24 25
```